

# Arishya Jindal

+65 82143951 | [arishya\\_jindal@mymail.sutd.edu.sg](mailto:arishya_jindal@mymail.sutd.edu.sg) | <https://arishya7.github.io/portfolio/>

Computer Science student building AI-driven automation, data pipelines, and full-stack applications that solve real-world problems

---

## EDUCATION

**Singapore University of Technology and Design** | B.Eng, Computer Science & Design (AI Minor) Sep 2023 - Apr 2027

Scholarship: AT-Capital Sustainability Scholarship

Awards: Youth Development for Climate Tech - 1st Place; Student Service Excellence Award (2025); Sustainability Spirit Award (2025); Orientation Excellence Award (2025); Singtel Information Systems & Programming Award - 3rd Place (2025)

---

## WORK EXPERIENCE

**System Requirements Engineer Intern** | Aumovio - Singapore May 2026 - Present

- Owned end-to-end automation of requirement extraction, verification, and change control across 75 requirement modules and 4 parameter modules, designing tools that verified AI-generated output to 100% accuracy before import.
- Led design of a Compare-and-Update pipeline synchronizing hundreds of verified requirement changes into IBM DOORS Next Generation (DNG), eliminating manual re-entry while preserving traceability and record integrity across Requirements Engineers in Singapore and India.
- Built a test suite and rule set from scratch, establishing edge-case coverage where no prior framework existed.
- Automated recurring workflows via MCP integration, cutting process runtime from 1-2 hours to 30 minutes.

**AI & Automation Intern** | Mummy's Market Pte Ltd - Singapore Aug 2025 - Jan 2026

- Built NLP pipelines (spaCy, KeyBERT, NLTK) to analyze large-scale consumer conversations and extract behavioral insights.
- Developed end-to-end Python data pipelines and SQL-backed analytics dashboards (Python, FastAPI, JavaScript) processing data from 500+ online sources, working with business teams to translate requirements into scalable solutions.
- Built Gemini-powered data validation and quality-control workflows, improving dataset reliability and reducing manual effort by 70%.

**Financial Technology, Extern** | RevRoad - Remote Jun 2024 - Jul 2024

- Analyzed five-year company performance datasets (Python, Matplotlib) and benchmarked 100+ EdTech business models to evaluate investment suitability and growth potential.

---

## ACADEMIC PROJECTS

**CareTech - Wearable Knee Device for Elderly Rehabilitation** | Capstone Project Feb 2026 - Present

- Shaped device design and clinical workflows by leading user research across CGH, SGH, and elderly care centers, interviewing 120+ stakeholders including patients, caregivers, and clinicians.
- Built an ML-based calibration pipeline to correct sensor drift, improving the accuracy and reliability of movement-tracking data used for clinical rehabilitation decisions.

**CDS DataBusters - Bus Service Analysis Using Multi-Source Data** | Academic Project Feb 2026 - Apr 2026

- Reduced bus wait-time prediction error by 72% (MAE: 1.26 min) by engineering contextual features from 5 live LTA DataMall endpoints (traffic incidents, speed bands, road works, faulty traffic lights) into a LightGBM prediction model.
- Built the Singapore Bus Intelligence Dashboard, an interactive 4-tab platform visualizing model predictions, sentiment, and reliability metrics across 559 bus services.

**CoolRide - YDCT Programme with SL2 x Meta** | Competition Programme Sep 2025 - Mar 2026

- Designed and built a climate-aware navigation system enabling cyclists to compare fastest vs. coolest routes, improving route comfort by 34% through shade-aware geospatial analysis - awarded 1st place.
- Deployed an interactive web app and Llama-powered Telegram chatbot on Hugging Face, using Supabase as backend to support real-time geospatial queries and route generation.

**Hotel Booking System - Web App Integrated with Ascenda APIs** | Academic Project May 2025 - Aug 2025

- Created system architecture and application workflows using 10+ UML diagrams (use case, class, sequence) to model business logic, user interactions, and backend processes.
- Built a full-stack web application using React and Node.js, implementing REST APIs for booking management, user authentication, and data handling.
- Validated core functionality through unit, integration, and system testing, ensuring reliable end-to-end workflows and data integrity.

**Evaluating Explainability in Reasoning LLMs** | UROP Research Sep 2025 - Feb 2026

- Investigated the explainability and reasoning quality of LLMs in mental-health support scenarios, where transparency, consistency, and safety are critical for everyday use.
- Built an evaluation framework benchmarking 7 model families (Claude, GPT, Llama, Qwen, Grok, Gemini, DeepSeek), scoring responses across 9 dimensions including technical accuracy, factual consistency, coherence, tone, and empathy.

SKILLS & OTHER INFORMATION

**AI/ML & Data:** LLMs, Agentic AI, MCP, NLP (spaCy, KeyBERT, NLTK), Prompt Engineering, Scikit-learn, OpenCV  
**Programming & Web:** Python (Pandas, NumPy, Matplotlib), Java, JavaScript, React, Node.js, HTML./CSS  
**Data, Tools & Infrastructure:** SQL, MongoDB, AWS (EC2, S3), Docker, Git, CI/CD fundamentals, Hugging Face, Firebase, IBM DOORS DNG, JIRA, Figma, Streamlit, Power BI, Tableau, Excel  
**Languages:** Fluent in English and Hindi, Intermediate in French. **Interests:** Running and Cycling

CO-CURRICULAR ACTIVITIES

- Secretary | Greenprint - Sustainable Club

Feb 2024 - Aug 2025

  - Coordinated with the Sustainability Office and faculty to dismantle and recycle 110+ prototypes across two batch years, redirecting 240L of wood, metal, and cardboard back into recycle and reuse.
  - Initiated indoor plant and cooking workshops promoting a farm-to-table concept, driving increased student use of SUTD's community garden, and recycled 2 boxes of Ang Baos into lanterns for the Mid-Autumn Festival.
- Design Facilitator and Mentor | SteamUnity

Jun 2024 - Feb 2025

  - Mentored 7 secondary and polytechnic students through the Design Innovation process, guiding 5+ prototype iterations of a single-use plastic solution while managing team conflicts.
  - Coached the team to selection as 1 of only 5 groups out of 28 to present their solution to DPM Heng Swee Keat.